

The empirical recurrence for OEIS A251303: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A251303 records “Empirical recurrence of order 74”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 74, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 74 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A251303 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with no 2×2 subblock having the sum of its diagonal elements less than the maximum of its antidiagonal elements.”. It has offset 1 and begins

26664, 2986712, 338614056, 38343620608, 4336649865709, 490318888438553, 55435729766236658, 6267641628409

The entry states:

Empirical recurrence of order 74 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:15:09, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 74 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{74} c_i a(n-i),$$

c_1	256	c_{20}	−1624381487764333131310	c_{39}	−4962852762382873821648554142
c_2	−26144	c_{21}	−21330343704181934899656	c_{40}	5384196299089519902671861011
c_3	1575540	c_{22}	152234645329439415952181	c_{41}	−5723178342689379453514437053
c_4	−64761744	c_{23}	−688831423167013736902987	c_{42}	5887772858603652942267346764
c_5	1952967343	c_{24}	2503997047310398344437616	c_{43}	−5783513202354730987536497641
c_6	−45213070411	c_{25}	−7812941780666987481862919	c_{44}	5361915021261073018896453447
c_7	830385074306	c_{26}	21526441349010957225170355	c_{45}	−4651558874315655577049683839
c_8	−12410160135135	c_{27}	−53161168037704885942054331	c_{46}	3753958478572672917684289055
c_9	153985283434668	c_{28}	118732157637511758243213127	c_{47}	−2807691613770239783399672656
c_{10}	−1611717099014634	c_{29}	−241248983487441468705133711	c_{48}	1941437122917365742292755024
c_{11}	14408460930596911	c_{30}	447888261331487819382009882	c_{49}	−1239069876071235098109905405
c_{12}	−111068719410042897	c_{31}	−762530446234745726339180460	c_{50}	728951238310253366951697119
c_{13}	743260191194836862	c_{32}	1194803308183557573199263689	c_{51}	−394784226461891695763365918
c_{14}	−4334450625181659912	c_{33}	−1730409072208624490122300004	c_{52}	196510736316572357523027241
c_{15}	22032872302186456158	c_{34}	2329754724855807296851869590	c_{53}	−89715786559511139008079936
c_{16}	−97113165310562463559	c_{35}	−2939526224890263899429669474	c_{54}	37463561110610569948256320
c_{17}	365469534782398966667	c_{36}	3514079495371866907727376637	c_{55}	−14258349756342691345100854
c_{18}	−1128534874470967886729	c_{37}	−4034569698567256448713298608	c_{56}	4924655579151111427010039
c_{19}	2526330148266551925781	c_{38}	4511214506520915143613591508	c_{57}	−1536189304064347689284834

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 74, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $74 + 4$ terms removed returns order 74 as well, so $\deg q_{\min} = 74 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $74 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 74 that this sequence satisfies — and that family turns out to have exactly one member.

References

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